Homework 5

Conditioning, covariance, and CDF transformations

60 points; 11 questions. Each part has its own printed point value.

Due: November 12, 2026, at 10 p.m. Pacific. Final no-penalty grace cutoff: November 13, 2026, at 10 p.m. Pacific.

Handwrite your work on paper or a tablet, subject to approved accommodations. Submit one legible PDF in Gradescope and map its pages to all eleven questions. Show reasoning, support, and assumptions. Exact expressions are accepted unless stated otherwise. Write independently and disclose collaborators. Do not use AI tools to generate, solve, check, or rewrite assessed work.

Notation. PMF means probability mass function; CDF means cumulative distribution function. A complete PMF or density includes its support and is zero elsewhere; a complete CDF covers every real threshold. Indicator notation and piecewise formulas are equally valid. The abbreviation iid means independent and identically distributed.

Study sequence. Start Q1–3 and Q9 after November 4, Q4–5 and Q10 after November 6, and Q6–8 plus Q11 after November 9. These are suggested stages, not additional deadlines.

1. Conditioning on Y

4 points

For Questions 1 and 2, $f_{X,Y}(x,y) = \frac{1}{2} I_{\{0<x<y<2\}}$. Its marginals are $f_X(x) = (1 - x/2) I_{\{0<x<2\}}$ and $f_Y(y) = (y/2) I_{\{0<y<2\}}$.

(a) [2] For each fixed y with $0 < y < 2$, find $f_{X|Y}(x \mid y)$ and its domain in x. Verify that it integrates to 1. Explain how conditioning on $Y = y$ is defined in this continuous model, where $P(Y = y) = 0$.

(b) [2] Calculate $P(X < 1/2 \mid Y = 3/2)$ and $E[X \mid Y = 3/2]$.

2. Conditioning on X

4 points

Use the same triangular model as in Question 1.

(a) [2] For each fixed x with $0 < x < 2$, find $f_{Y|X}(y \mid x)$ and its domain in y. Calculate $E[Y \mid X = x]$ and check that this mean lies in the conditional range of Y.

(b) [2] Find $P(Y > 3/2 \mid X = 1/2)$. Compare the formulas and domains of $f_{Y|X}(y \mid x)$ and $f_{X|Y}(x \mid y)$ from Question 1, explaining the effect of changing which variable is conditioned on.

3. A conditional density on a square

4 points

Let $f_{R,S}(r,s) = (r+s) I_{\{0<r<1,\ 0<s<1\}}$.

(a) [2] Find $f_S(s)$ and $f_{R|S}(r \mid s)$. State the allowed values of s and, for each such s, the domain in r. Verify that the conditional density integrates to 1.

(b) [2] Find $E[R \mid S = s]$ for $0 < s < 1$. Are R and S independent? Justify your conclusion using their conditional distribution.

4. Covariance and variances of combinations 6 points

Use $f_{X,Y}(x,y) = \frac{1}{2} I_{\{0<x<y<2\}}$. You may use $E[X] = 2/3$, $E[Y] = 4/3$, and $\mathrm{Var}(X) = \mathrm{Var}(Y) = 2/9$.

(a) [3] Compute $E[XY]$ by double integration. Use it to find $\mathrm{Cov}(X,Y)$ and $\mathrm{Corr}(X,Y)$.

(b) [3] Find $\mathrm{Var}(Y - X)$ and $\mathrm{Var}(2X - Y + 3)$. Show both variance expansions and explain the contribution of covariance in each.

5. A variable and its square 4 points

Each value in $\{-2, -1, 0, 1, 2\}$ has probability $1/5$ for A. Define $B = A^2$.

(a) [2] Calculate $E[A]$, $E[B]$, and $E[AB]$, and use them to find $\mathrm{Cov}(A,B)$.

(b) [2] Are A and B independent? Justify your answer with a probability comparison. Find $\mathrm{Var}(A+B)$; you may use $E[A^4] = 34/5$.

For Questions 6–8, first derive the input CDF from its PDF, then follow the five lecture steps: **1.** output support; **2.** event algebra; **3.** probability written through the input CDF; **4.** complete output CDF; **5.** density and checks. Show the CDF expression before substituting its formula. When a one-to-one map has an inverse, use it to translate the event; justify the inequality direction. The inverse-density formula may be used as an extra check, but it does not replace the requested CDF derivation. No distribution names need to be memorized. Each part displays its points.

6. An inverse-square score 6 points

Context: a positive size measurement is converted to a bounded score; larger inputs give smaller scores.

Let R have density $f_R(r) = 3r^{-4}$ for $r > 1$, and 0 otherwise. Define $Y = 1/R^2$. No named-distribution formula is needed.

(a) [1] Input CDF. Integrate the given PDF to derive $F_R(r)$ for every real r.

(b) [1] Step 1: output support. Find the range of Y, using the behavior of $1/r^2$ at both ends of the input range.

(c) [1] Step 2: event. For $0 < y < 1$, rewrite $\{Y \le y\}$ as an event involving R alone. Show every inequality step.

(d) [1] Step 3: input CDF. Write $F_Y(y)$ explicitly in terms of F_R before evaluating it. Explain why an upper tail appears.

(e) [1] Step 4: complete CDF. State $F_Y(y)$ for all real y and check its limits at the output boundaries.

(f) [1] Step 5: density and check. Differentiate on the output range, state a density with zero elsewhere, and verify that its integral is 1.

7. Squaring an asymmetric error 10 points

Context: a signed error can range farther above zero than below it; squaring records its size without its sign.

Let V have density $f_V(v) = 2(v+1)/9$ for $-1 < v < 2$, and 0 otherwise. Define $Q = V^2$.

(a) [1] Input CDF. Integrate the PDF and give $F_V(v)$ for all real v.

(b) [1] Step 1: output support. Find the output range and identify every interior threshold at which the contributing input interval changes form.

(c) [2] Step 2: event. For each interior regime from part (b), translate $\{V^2 \le q\}$ and intersect the resulting interval with $-1 < V < 2$. Show both branches before clipping.

(d) [2] Step 3: input CDF. In each interior regime, write $F_Q(q)$ as a difference of values of F_V, then substitute and simplify.

(e) [2] Step 4: complete CDF. Give $F_Q(q)$ for every real q. Check its boundary values and continuity at the interior threshold. Is there an atom at that threshold?

(f) [2] Step 5: density and check. Find a density of Q on each open output interval, set it to zero elsewhere, and verify normalization by integrating both pieces.

8. From a fraction to a ratio 6 points

Context: a load fraction is converted to the ratio of used capacity to spare capacity.

Let X have density $f_X(x) = 6x(1-x)$ for $0 < x < 1$, and 0 otherwise. Define $T = X/(1-X)$.

(a) [1] Input CDF. Derive $F_X(x)$ from the PDF, including its exterior branches.

(b) [1] Step 1: output support. Find the range of T and justify it with monotonicity and endpoint limits.

(c) [1] Step 2: event. For $t > 0$, solve $X/(1-X) \le t$ for X. State why each multiplication or division preserves the inequality.

(d) [1] Step 3: input CDF. Write $F_T(t)$ explicitly in terms of F_X, then substitute the input CDF and simplify.

(e) [1] Step 4: complete CDF. State $F_T(t)$ for all real t and check its limits at 0 and $+\infty$.

(f) [1] Step 5: density and check. Differentiate the CDF, state a density with zero elsewhere, and verify normalization. For the integral, the substitution $w = 1 + t$ may help.

9. Conditioning in a joint table 4 points

The joint PMF of (A, B) is $p(0,0) = 1/10$, $p(0,1) = 2/10$, $p(1,0) = 3/10$, and $p(1,1) = 4/10$, with zero probability elsewhere.

(a) [2] Find the complete conditional PMF of A given $B = 1$, and calculate $E[A \mid B = 1]$.

(b) [2] Find $E[A \mid B = 0]$. Are A and B independent? Give a decisive probability comparison.

10. Variability of a sum and an average 4 points

$X_1, X_2, \ldots$ are iid with mean 4 and variance 4. Write $S_n = \sum_{i=1}^{n} X_i$ and $\overline{X}_n = S_n/n$.

(a) [2] Derive the mean and variance of S_n and $\overline{X}_n$. Identify which step uses independence.

(b) [2] Find the standard deviation of $\overline{X}_n$ for $n = 25$ and $n = 100$. Explain what changes when the sample size quadruples and whether the standard deviation of an individual X_i changes.

11. The probability integral transform 8 points

Context: a continuous observation can be converted to its percentile; an inverse CDF can turn a uniform input into a desired distribution.

Let X have a continuous CDF F_X, and set $U = F_X(X)$. Prove that $U \sim \text{Uniform}(0,1)$ by deriving its CDF. Continuity alone need not make F_X strictly increasing.

(a) [3] First, an ordinary inverse. Assume $X \in I$ with probability 1 for an open interval I, and $F_X : I \to (0,1)$ is continuous and strictly increasing onto $(0,1)$. For $0 < u < 1$, rewrite $\{U \le u\}$ using $F_X^{-1}(u)$, express $F_U(u)$ in terms of F_X, and evaluate. Give the complete CDF of U, including $u = 0, 1$, and identify its distribution.

(b) [3] Now, any continuous CDF. Drop the strict-increase assumption. For $0 < u < 1$, define the rightmost cutoff at CDF level u by

$$b_u = \sup\{x \in \mathbb{R} : F_X(x) \le u\}.$$

Here sup means the least upper bound. You may use this fact: continuity and the CDF limits ensure that b_u is finite and $F_X(b_u) = u$. Show both directions of $\{F_X(X) \le u\} = \{X \le b_u\}$. Use this event equality to obtain $F_U(u)$, finish the exterior branches, and explain why flat portions of F_X cause no problem.

(c) [2] The inverse construction. Return to the assumptions in (a). Let $A \sim \text{Uniform}(0,1)$ and define $Z = F_X^{-1}(A)$. State the output range. Derive $F_Z(x)$ using F_A and F_X, with the exterior cases included, and conclude that Z has the same distribution as X.

Reading references. Week 6 notes on conditioning and covariance; Week 7 notes on the CDF method and one-to-one transformations. Question 11 is a new guided application. Harchol-Balter: pp. 157 and 161, pp. 88–92, and Exercise 7.10. DeGroot–Schervish: Sections 3.6 and 3.8, 4.6–4.7. Consult selected passages as needed; neither whole chapters nor both books are required.